NEXUS ANALYTICS

Implementation & Verification Status
Build 0.2.0 · 10 November 2026

1. Delivered application capabilities

Business onboarding, password hashing and signed sessions; business-scoped data access; workspace creation; CSV and bundled synthetic-data ingestion; SQL analytics; model metrics and inference; interactive charts; Snwolley-assisted management reporting; PDF report generation; and updated setup/architecture documentation.

2. Automated verification

Check	Result
Python compilation	Passed
Unit/API tests	14 passed
PostgreSQL integration tests	5 skipped — no PostgreSQL server available
Model training	Completed
Saved-model inference API	Passed
Password/session tests	Passed
AI prompt and response tests	Passed with mocked provider
AI PDF response	Valid PDF with mocked provider
TypeScript/JSX syntax check	Passed with temporary declaration shims
Full Next.js production build	Not run — dependencies/build binary unavailable

3. Live integration checks

PostgreSQL could not be started in the verification environment: Docker Engine and the PostgreSQL server binary were unavailable, and localhost:5432 refused connections. Schema initialisation, migration execution, database COPY ingestion and live analytical SQL therefore remain unverified. Snwolley could not be called live because no current valid API key was configured. The previously shared testing credential was deliberately excluded from the final project; obtain a fresh key and place it only in backend/.env.

4. Model metrics

Metric	Value
ROC AUC	0.6268
Average precision	0.0320
Threshold	0.1267
Precision	4.08%
Recall	26.45%
F1	0.0707

5. Final acceptance required

On the developer machine: configure a fresh session secret and Snwolley key; start PostgreSQL with Docker Compose; run schema initialisation; register a business; load 30,000 sample rows; verify dashboard analytics and model prediction; generate an AI report/PDF; run all tests with PostgreSQL enabled; run npm install and npm run build; and execute the

opt-in live integration script. Full acceptance is not claimed until these checks succeed.